Instruction Manual

Magnetometry with spins in diamond

PHY405: Condensed Matter Physics Lab



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1 Safety instructions

1. Laser radiation can cause injury through thermal exposure; avoid looking directly into the laser beam.
2. While handling optics in the setup, avoid touching the surfaces with your fingers to prevent oil and dust contamination, which can affect optical performance and potentially damage the equipment.
3. Avoid touching the current carrying Helmholtz coils, as this can result in electric shocks or other injuries.
4. Do not disturb the diamond placed on the PCB (printed circuit board), as it may change its orientation with respect to the magnetic field generated by the coils.
5. Ensuring the proper alignment of the laser is crucial for its safe and effective operation. Avoid any adjustments to the laser alignment.



Caution : Laser radiation predominantly causes injury via thermal effects; avoid looking directly into the laser beam.

2 Learning goals

The experiment introduces Zeeman effect measurements using nitrogen-vacancy (NV) centers in diamond. You will apply the concepts of quantum mechanics and solid-state physics in this experiment and will also gain hands-on experience with magnetic field measurements. The learning objectives are listed below.

1. Learn to operate a diamond magnetometer and perform optically detected electron-spin resonance experiments.
2. Learn to use two axis Helmholtz coil setup to control the direction and magnitude of the magnetic fields in the experiment.
3. Determine the projection of an external magnetic field on each NV center orientation using the experimental data.

3 The nitrogen-vacancy center in diamond

The negatively charged nitrogen-vacancy defect, abbreviated as NV center, is a paramagnetic spin-1 defect in the diamond. The NV axis, along with the vacancy's nearest neighbor carbon atoms, form a tetrahedron (see fig. 1(a)) and the symmetry axis that connects the nitrogen atom to the vacancy is called the NV axis or NV orientation. The NV center in diamond possesses two unpaired electrons, resulting in a spin-1 system. The spin states of the NV defect can be described by the eigenstates of the $\hat{S}_z$ operator, where the quantization axis z is along the NV axis. In the ground state of the NV center, the spin sublevel $|m_s = 0\rangle$ and the degenerate states $|m_s = \pm 1\rangle$ are separated by the zero-field splitting denoted by D ($\approx$ 2.870 GHz). The zero-field splitting arises from the dipolar spin-spin interaction. When an external magnetic field is applied, the degeneracy of the states $|m_s = \pm 1\rangle$ is lifted due to the Zeeman effect.

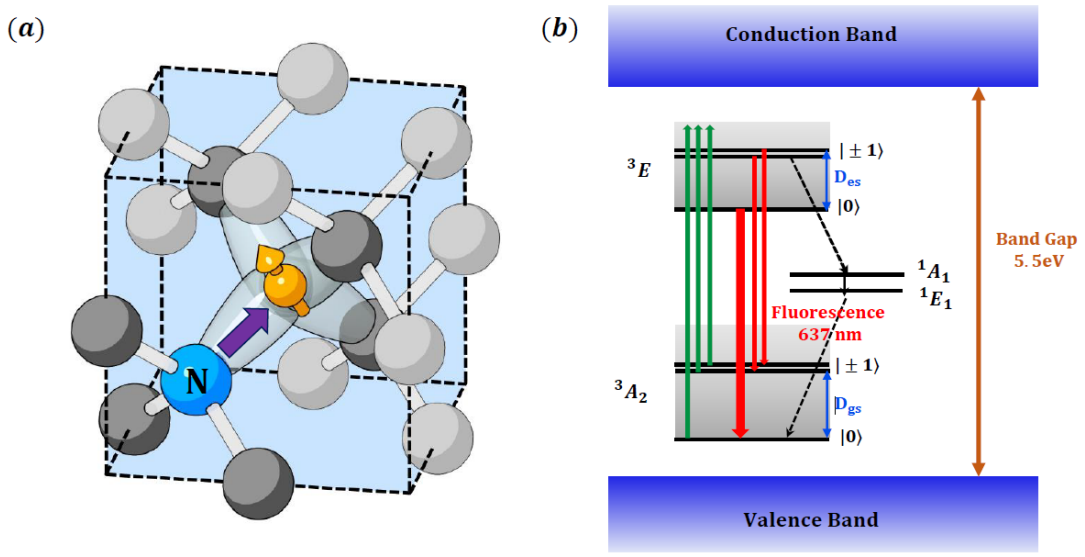


Figure 1: **(a)** The NV center consists of a substitutional nitrogen atom adjacent to a vacancy. Nitrogen atom in blue, vacancy in light blue, and the nearest neighbour carbon atoms surrounding the NV in dark grey. The electron spin of the NV center is depicted in yellow. NV orientation is depicted by a purple color arrow. (Image Ref. [1]) **(b)** Energy level diagram of the NV center in diamond.

A green laser can be used to excite the NV center in the diamond from its electronic ground state $^3A_2$ to its electronic excited state ^3E as shown in fig. 1(b). If the electron is initially in $|m_s = 0\rangle$ sublevel of its electronic ground state $^3A_2$, it will be excited to the $|m_s = 0\rangle$ sublevel of the electronic excited state ^3E. The NV center in this excited state has a lifetime of around a few nanoseconds and will relax to the same spin sub-level of the electronic ground state because of the spin conservation [2]. This process is accompanied by the emission of a red photon in the wavelength range of 650-800 nm. On the other hand, if the electron is initially in the $|m_s = \pm 1\rangle$ sub-level of the electronic ground state $^3A_2$, it will be excited to the $|m_s = \pm 1\rangle$ sublevel of the electronic excited state ^3E. In this case, while relaxing to the ground state, there are two transition pathways: the electron in this excited state can relax to the $|m_s = \pm 1\rangle$ sub-level of the ground state $^3A_2$ (spin-conserving transition), or the excited state can relax to the $|m_s = 0\rangle$ sublevel of the ground state $^3A_2$ via an intermediate metastable state (long-lived singlet state). This transition into a singlet state is a non-radiative decay pathway, also known as 'intersystem crossing (ISC)' (see fig. 1(b)) [3], which results in a fluorescence contrast of approximately 30% between the $m_s = 0$ and $m_s = \pm 1$ states [4]. This is why $|m_s = 0\rangle$ and $|m_s = \pm 1\rangle$ are also termed as 'bright' and 'dark' states, respectively and their fluorescence emission intensity can be used to perform optical readout of the spin states. Additionally, the ISC transition pathway

also leads to the initialization of the NV center spin state to $|m_s = 0\rangle$ under continuous optical excitation.

Zeeman splitting of NV spin states

Let us now find the energy levels of the ground state spin-triplet under an external magnetic field using quantum mechanical formalism. For small DC magnetic fields, the ground state spin Hamiltonian of NV center is given by

$$\hat{H}_0 = D\hat{S}_z^2 + \gamma B_z \hat{S}_z \tag{1}$$

where $\hat{S}_z$ is the z component of the spin-1 vector operator $\hat{\boldsymbol{S}}$, γ (=28 MHz/mT) is the gyromagnetic ratio for the NV center electron spin and B_z is the component of the external magnetic field along the NV quantization axis. Note that we have taken $\hbar$=1 throughout the manual. The matrix representation for $\hat{S}_z$ in the $\hat{S}_z$ basis $\{|+1\rangle, |0\rangle, |-1\rangle\}$ is given by

$$\hat{S}_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

Solving the above Hamiltonian, we get the eigenenergies of the NV center spin in the presence of an external magnetic field as

$$E_{|+1\rangle} = D + \gamma B_z, \tag{2a}$$
$$E_{|-1\rangle} = D - \gamma B_z, \tag{2b}$$
$$E_{|0\rangle} = 0. \tag{2c}$$

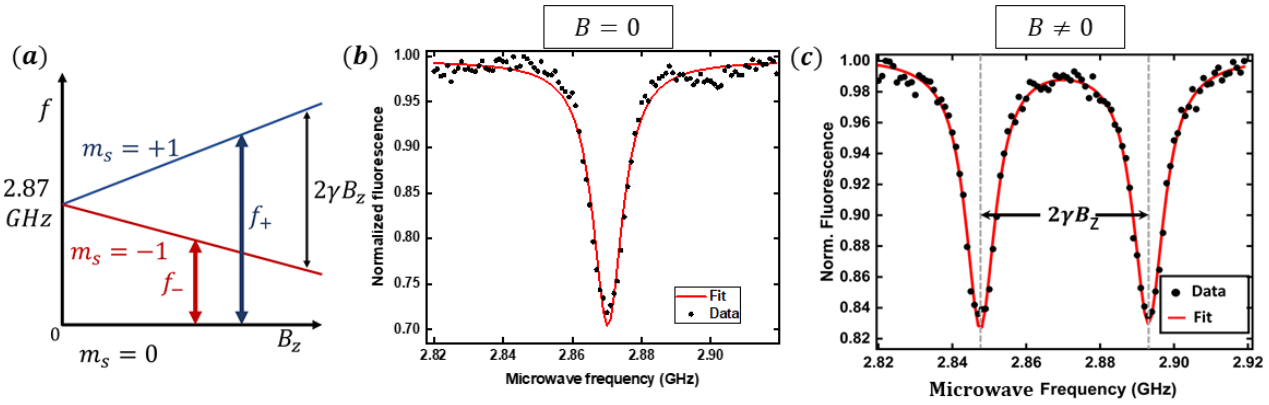


Figure 2: **(a)** Energy level diagram of NV center under an external magnetic field. **(b)** Experimental ODESR spectrum at zero magnetic field. **(c)** Experimental ODESR spectrum at finite magnetic field. The splitting between the two dips gives the component of the external magnetic field along the NV axis.

Optically detected electron spin resonance

As explained in the previous section, the Zeeman splitting between the states $|m_s = +1\rangle$ and $|m_s = -1\rangle$ varies linearly with the magnetic field component B_z parallel to the NV axis. A resonant microwave (MW) field [1] [5] can be applied to induce the spin transition $|m_s = 0\rangle \rightarrow$

[1] **Note.** In this manual, "MW field" refers to the magnetic field component of the microwave.

$|m_s = -1\rangle$ or $|m_s = 0\rangle \rightarrow |m_s = +1\rangle$ with frequencies f_- and f_+, respectively as illustrated in fig. 2(a). These resonant spin transitions can be detected optically due to the reduction of the NV fluorescence rate resulting from the transfer of the population between the spin states. Hence, by continuously sweeping the applied MW frequency and simultaneously collecting the NV fluorescence, we can obtain a spectrum similar to fig. 2(b) at zero magnetic field and fig. 2(c) at finite magnetic field for a single NV center. This technique is known as optically detected electron spin resonance (ODESR) or optically detected magnetic resonance (ODMR) [3].

4 Ensemble of NV centers

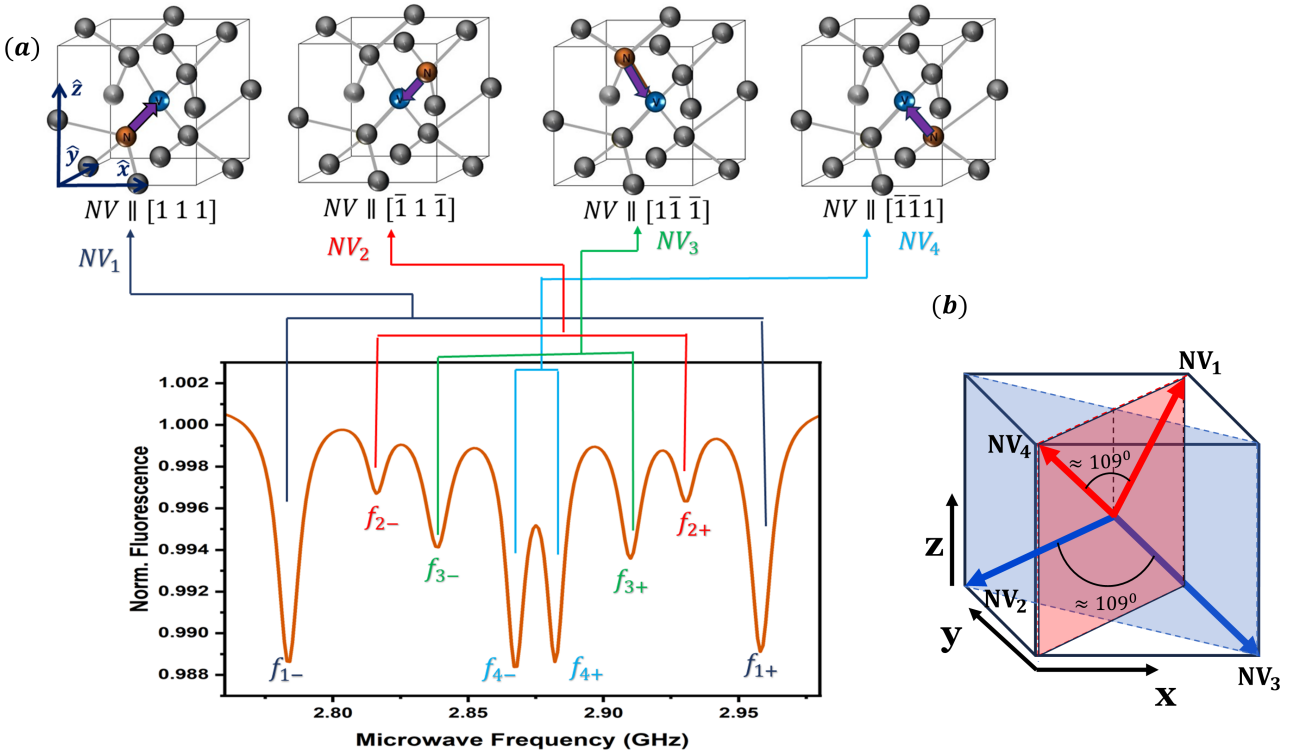


Figure 3: **(a)** (100)-oriented diamond have four possible NV orientations ([1, 1, 1], [1̄, 1, 1̄], [1, 1̄, 1̄], and [1̄, 1̄, 1]). ODMR spectrum represents pair of peaks that correspond to NV orientation. **(b)** Composite representation of all 4 possible orientations with considering nitrogen as the origin of coordinate system.

In this section, we expand the theoretical description to diamonds containing dense ensembles of NV centers. This necessitates considering all possible NV orientations within the diamond crystal. Due to the tetrahedral nature of the diamond crystal lattice (fig. 1(a)), there are four possible NV orientations in the diamond lattice as shown in fig. 3(a). Thus, we can define each NV orientation in the diamond's frame as [1, 1, 1], [1̄, 1, 1̄], [1, 1̄, 1̄], and [1̄, 1̄, 1] [6], and these NV orientations can be assigned as NV_1, NV_2, NV_3, and NV_4 respectively (see fig. 3(b)). Also, note that NV_1 and NV_4 are present in [1̄10] plane while NV_3 and NV_2 are present in [110] plane, and both planes are normal to each other. All the four orientations of NV centers can be represented by four different unit vectors.

$$\hat{\mathbf{u}}_{\mathbf{NV_1}} = \frac{1}{\sqrt{3}}(1,1,1) \qquad \hat{\mathbf{u}}_{\mathbf{NV_2}} = \frac{1}{\sqrt{3}}(-1,1,-1)$$

$$\hat{\mathbf{u}}_{\mathbf{NV_3}} = \frac{1}{\sqrt{3}}(1,-1,-1) \qquad \hat{\mathbf{u}}_{\mathbf{NV_4}} = \frac{1}{\sqrt{3}}(-1,-1,1)$$

As discussed earlier, the observed Zeeman splitting in ODMR spectrum, is proportional to the magnetic field projection along the NV center axis. In NV ensembles, the projection of an external magnetic field $\mathbf{B} = (B_x, B_y, B_z)$ on each NV orientation is given by

$$B_{NV_1} = \mathbf{B}.\hat{\mathbf{u}}_{\mathbf{NV_1}} = \frac{1}{\sqrt{3}}(B_x + B_y + B_z) \tag{3a}$$

$$B_{NV_2} = \mathbf{B}.\hat{\mathbf{u}}_{\mathbf{NV_2}} = \frac{1}{\sqrt{3}}(-B_x + B_y - B_z) \tag{3b}$$

$$B_{NV_3} = \mathbf{B}.\hat{\mathbf{u}}_{\mathbf{NV_3}} = \frac{1}{\sqrt{3}}(B_x - B_y - B_z) \tag{3c}$$

$$B_{NV_4} = \mathbf{B}.\hat{\mathbf{u}}_{\mathbf{NV_4}} = \frac{1}{\sqrt{3}}(-B_x - B_y + B_z) \tag{3d}$$

Here, B_x, B_y and B_z are represented in the diamond frame.[2] Thus, the number of dips observed in the ODMR spectrum depends on the orientation of the magnetic field with respect to the crystallographic axes. In general, it is possible to observe eight dips in the ODMR spectra for an arbitrary orientation of the magnetic field. However, the number of dips observed in the ODMR spectra is reduced for certain directions of the magnetic field owing to symmetry considerations. This is explained in section 6.

5 Experimental setup

This section provides a description of our experimental setup and the various optical and electronic components necessary for its operation. Fig. 4 shows the schematic of the experimental setup. The diamond used in this experiment is a high pressure high temperature (HPHT) grown diamond having a large density of NV centers. When the diamond is illuminated with a green laser, due to the high density of NV centers it emits red fluorescence which is even visible to the naked eye.

A fiber-coupled green laser source is connected to a fiber port which outputs a collimated laser beam. It is then directed towards the dichroic mirror, which reflects the green laser beam towards the objective lens. This objective lens focuses the beam on the diamond sample (see fig 4), thus optically exciting the NV centers, resulting in the emission of red fluorescence in all directions. The objective lens captures a small portion of the emitted fluorescence, which first gets transmitted through the dichroic mirror and then the long-pass filter. The long-pass filter blocks the green laser beam, thus eliminating the background from the red fluorescence signal. This filtered beam is then focused on to the photodiode sensor using a condenser lens. The photodiode converts incident fluorescence into an analog voltage, which is then acquired by the data acquisition hardware (DAQ). Finally, the DAQ signals can be monitored and analyzed using LabVIEW software on the computer.

[2]**Note.** Throughout this manual, (X, Y, Z) refers to the lab frame axes and (x, y, z) refers to the diamond's frame axes. All the analysis is done in diamond's frame.

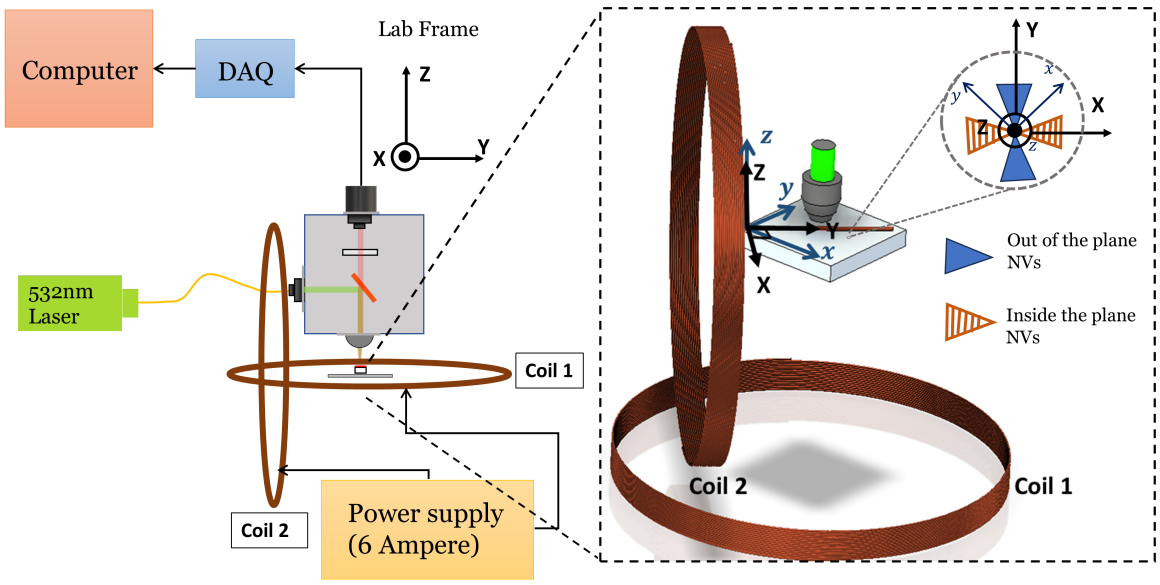


Figure 4: Schematic of the experimental setup showing both the optics and electronics used in the setup. In our setup, the diamond's frame (x, y, z) is rotated by (45°) with respect to lab frame (X, Y, Z). The whole experimental analysis is done in the diamond frame. Note, how the microwave wire is placed on the top of the diamond surface.

In the experimental setup, as shown in fig. 5, the diamond sample is glued on a printed circuit board (PCB) which is specially designed for the delivery of the microwave signals to the diamond. It is to be noted that the diamond is glued such that its edges are rotated by 45° with respect to the lab frame, which we denote by (X, Y, Z). Due to this, the diamond's frame, here we denote it by (x, y, z), is also rotated by 45° with respect to the lab frame (Fig. 4). In this experiment all the analysis is done in the diamond's frame. The PCB (holding the diamond) is fixed to a translation stage which can move it in all the three directions, while the objective lens remains fixed. In this setup air objective lens is used. To deliver the microwave to the NV spins a 20-micron copper wire is placed on top of the diamond sample's surface. This microwave is used to drive the NV spins from the ground spin-state m_s=0 to the ground excited state, m_s=±1 state (fig. 1(b)). This MW signal is generated by a signal generator which is passed through the copper wire after being amplified by a MW amplifier. To apply the magnetic field to the NVs, two circular coils of 400 mm and 150 turns are used. Each of these coils are connected to a separate power supply (of 30V/5A rating) in order to individually control the magnitude of the applied magnetic field. As shown in the fig. 4 the two coils are placed perpendicular to each other, coil 1 is placed along lab frames Y axis and coil 2 along lab frames Z axis. Thus, by varying the magnitude of the magnetic field generated by the two coils one can control the direction of total applied magnetic field.

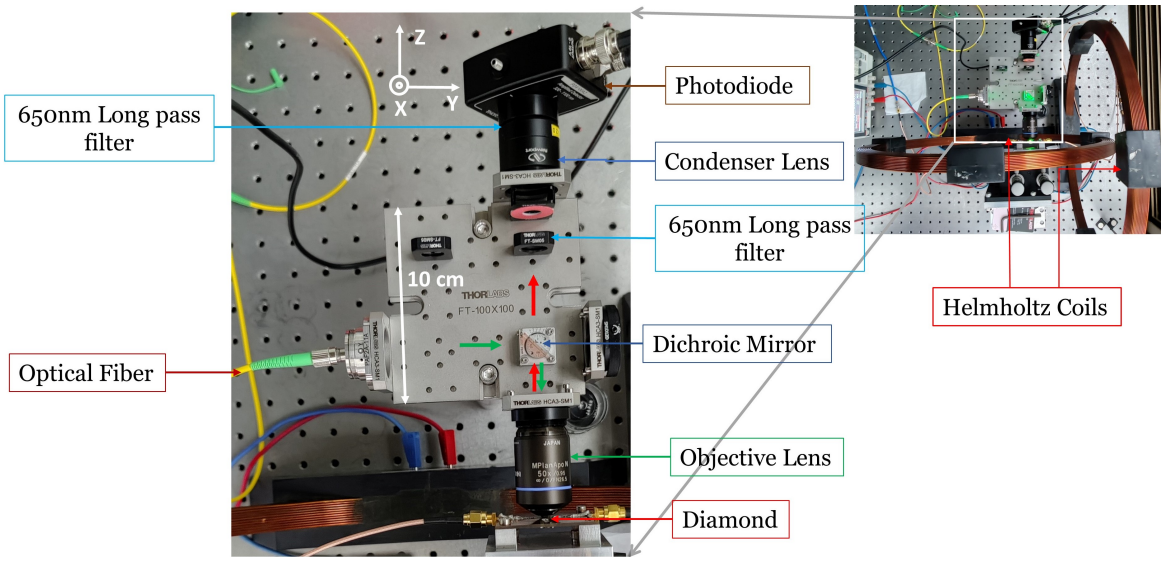


Figure 5: Diamond Magnetometer. Green arrows denote the laser excitation path, while Red arrows denote the fluorescence detection path.

6 Theoretical framework of the experiment:

Here we discuss the theoretical background necessary for understanding the experimental ODMR spectra, i.e, how does the ODMR spectrum varies for different magnetic field directions with respect to the four NV orientations.

As we discussed earlier, the number of dips observed in the ODMR spectrum depends on the projection of the applied magnetic field on each of the NV orientations. In this section, we discuss various cases of magnetic field that give rise to a distinct number of dips in the ODMR spectrum.

6.1 Case 1: condition for one peak

If the projection of the magnetic field on each NV orientation is zero, then we observe only one dip in the ODMR spectrum (fig. 6). This is possible only when there is no applied magnetic field (B=0) and the dip is observed at 2.87 GHz corresponding to zero-field splitting (refer Sec.3). In the experiment one may also observe a small splitting within this dip. It is due to the electric and strain fields present in the lattice.

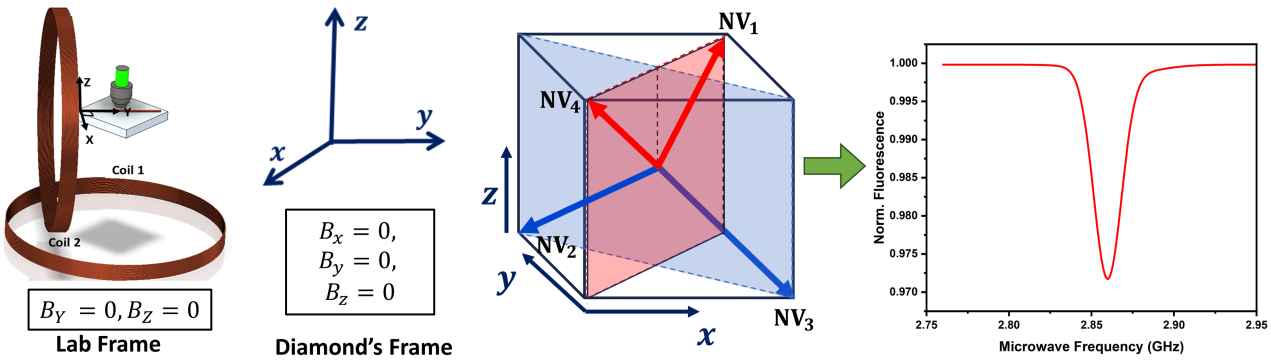


Figure 6: When $B_{NV_1} = B_{NV_2} = B_{NV_3} = B_{NV_4} = 0$, we observe one peak in the ODMR spectrum.

6.2 Case 2: condition for two peaks

When the magnetic field projection on each NV orientation is equal, we observe only two dips in the ODMR spectrum. It is because the Zeeman splitting for each of the NV will be the same. This can be achieved by applying the magnetic field normal to the surface of the diamond. The configuration of our experimental setup allows us to apply the magnetic field along the z-axis of the diamond as shown in the figure below.

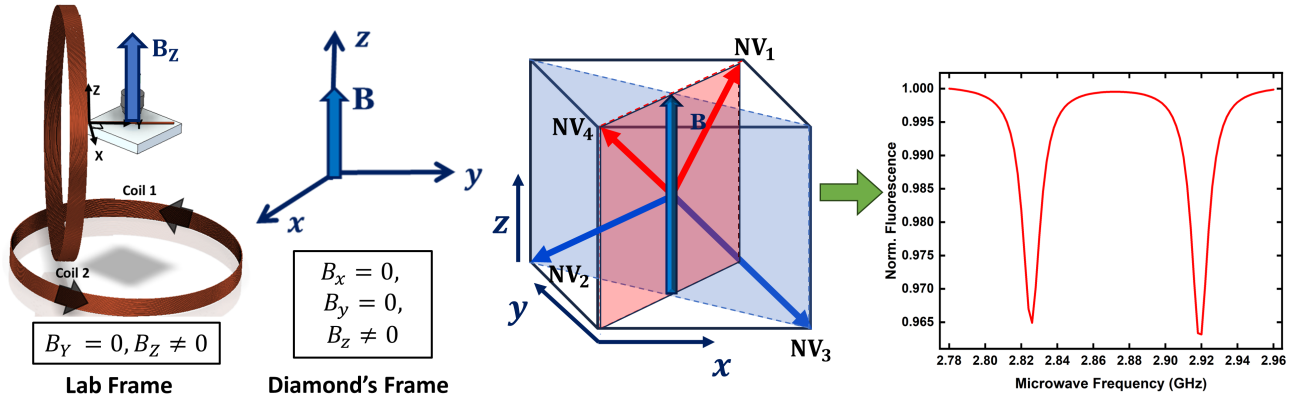


Figure 7: When $B_{NV_1} = B_{NV_2} = B_{NV_3} = B_{NV_4} \neq 0$, then we observe only two dips in the ODMR spectrum.

6.3 Case 3: condition for three peaks

When the magnetic field is applied along the 110 direction of the diamond, the field is normal to the plane containing the two NV centers (say, NV_2 and NV_3; fig. 8). Thus, the projections on these two NV orientations will be zero. While, the magnetic field lies in the plane containing the other two NV centers (NV_1 and NV_4 as shown in fig. 8) and has an equal projection on NV_1 and NV_4. As a result, we observe three dips in the ODMR spectrum. The middle peak corresponds to the NV_2 and NV_3 (there is Zeeman splitting), while the outer two peaks correspond to the NV_1 and NV_4. Since our diamond is placed at $\approx 45^0$ degree with respect to the lab frame, the projections of the magnetic field along the x and y axes of the diamond's frame or crystal frame are equal to each other.

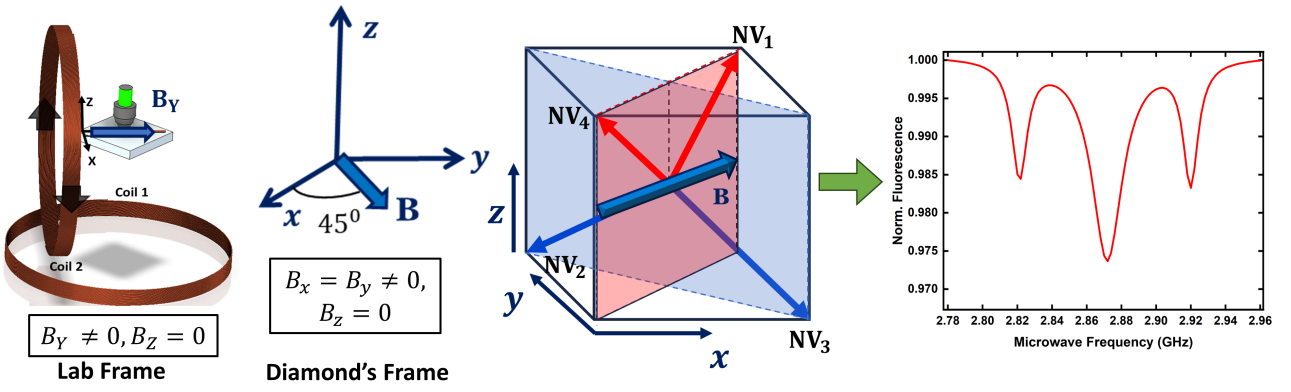


Figure 8: Case 3: When $B_{NV_1} = B_{NV_4} \neq 0$ and $B_{NV_2} = B_{NV_3} = 0$, then we observe three peaks in the ODMR spectrum.

6.4 Case 4: condition for four peaks

When the magnetic field aligns with one of the NV orientation (say, NV_1, fig. 9), it results in a larger projection on that NV, and thus a large Zeeman splitting. While the other three NV orientations (NV_2, NV_3 and NV_4, fig. 9) exhibit equal projections, hence they will have the same Zeeman splitting. This results in four peaks in the ODMR spectrum (fig. 9). The outer two corresponds to the NV_1 and the inner two peaks to the NV_2, NV_3 and NV_4. To achieve this in the experiment, it is required to tune the magnetic field of the coils along the Y-axis and Z-axis such that $B_x = B_y = B_z \neq 0$.

Question: So, there is one more way in which one can apply the magnetic field to obtain the four dips in the ODMR spectrum. Can you think of that condition? If you can, then is it possible to obtain that condition with the present setup? Hint: There are two NV pairs, and adjusting the magnetic field for them so that the two pairs experience different field projection (but same within the pair), might be the key.

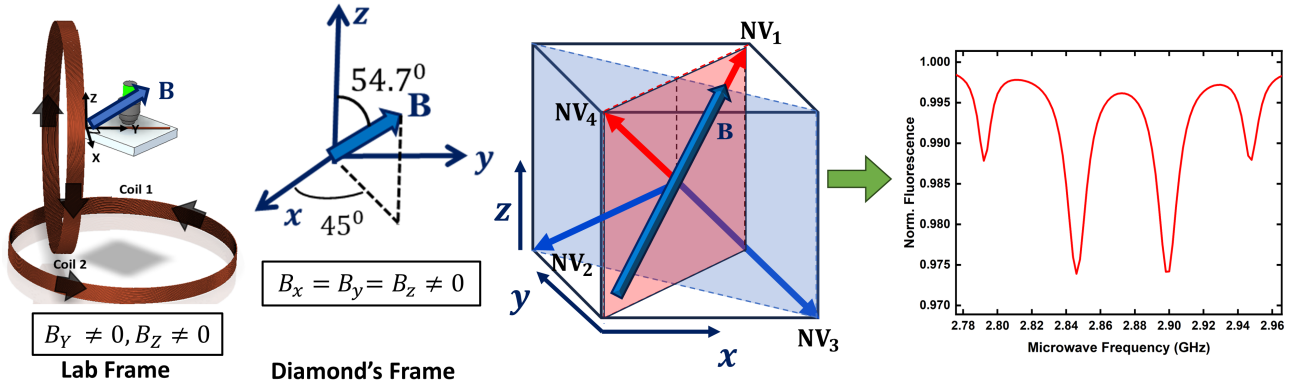


Figure 9: Case 4: When $B_{NV_2} = B_{NV_3} = B_{NV_4} \neq B_{NV_1} \neq 0$, then we observe four peaks in the ODMR spectrum.

For more detailed information about the nitrogen-vacancy center, refer to Suter's review paper [7]. It provides comprehensive insights into this topic, covering various aspects of the nitrogen-vacancy center and its applications in quantum sensing.

7 Lab report

1. Make a list of the various components used in the experimental setup and explain their functionality.
2. How does the projection of the magnetic field on different NV orientations influence the number of dips in the ODMR spectra?
3. What is the role of the microwave field to obtain the ODMR spectrum? What is the underlying physics behind it?
4. Make a note of the current (I) values used for the experiment in the table below. For the given current value and coil specifications, find the magnetic field vector in the diamond's frame using the formula below:

   Magnetic field produced by the coils:

$$|B| = \frac{\mu_0 n I}{2\pi} \frac{R^2}{(R^2 + Z^2)^{\frac{3}{2}}}$$

where n is the number of turns, I is the current, R is the radius of the coil, and Z is the axial distance from the coil.

The magnetic field along each NV orientation can be found using the Zeeman effect formula.
Formula:

$$\Delta\nu_i = 2\gamma_{NV} B_i$$

where, $\Delta\nu_i$ is the difference between the two peaks symmetric about the ~2.87 GHz, γ_{NV} =28 MHz/mT and B_i = magnetic field projection along the i_{th} NV.

No. of Peaks	Current, I_Y	Current, I_Z	Diamond Frame, $B = (B_x, B_y, B_z)$	B_1 (mT)	B_2 (mT)	B_3 (mT)	B_4 (mT)

To Do: Prepare a table for each of the cases mentioned in sec. 6, and do the following:

(a) Conduct the experiment using at least three distinct current values while maintaining a consistent number of peaks in the ODMR spectrum, such that the splitting increases. Except for the Case 1 (of course) and Case 4.

(b) In Case 4, obtain the ODMR spectrum for only one set of I_y and I_z values and fill the table.

(c) As mentioned in point number 4 above, using the specification of the coils (no. of turns =150; diameter = 400mm) and knowing the approximate distance of the diamond from the two coils, calculate B = (B_x, B_y, B_z). You may take help of any software (e.g. Mathematica, python etc.). Stick the print-out of the code used.

(d) Then using the Zeeman splitting formula (which is mentioned above the table), find the magnetic field that each of the NV would be experiencing. Do this for all the distinct current values you used to obtain the ODMR spectrum.

***Kindly pay attention to the units of the physical quantities.**

5. Now that you have performed this experiment, in what ways do you think NV centers can expand our understanding of physics? Also, what potential applications do you envision for the NV centers in diamond?

8 Appendix

8.1 Technical instructions

1. Make sure the DAQ (data acquisition device) is ON.
2. Switch on the microwave signal generator first and then the microwave amplifier.

3. Switch on the laser and set the laser power at its maximum value.
4. Switch on the photo-diode and check if the the gain value is set at 0 dB.
5. If the alignment is proper, then the laser beam will be incident on the diamond surface and one can see bright collimated red fluorescence after the long-pass filter. One can see this fluorescence using a white screen (any rough paper or white card).
6. Now, we are ready to perform the experiment. For that we open the Labview 2017 program on the PC as instructed below.

8.2 LabVIEW instructions

Please follow the instructions below for operating the LabVIEW program.

1. Open the custom-developed LabVIEW program which is to be used for performing the experiment: **CDM_NV_magnetometry.vi**
2. Before running the program, enter the values in **SWEEP CONTROL PANEL**:
   - First set frequency range by entering the value in the **Start frequency (GHz)** and **End frequency (GHz)**. You should set the range such that ODMR spectrum falls within this range of MW frequency.
   - Set the microwave power by entering a value in **RF Amplitude**, as instructed by the TA.
   - Set the increment/step of the MW frequency by entering a value in **Step frequency (MHz)** . Optimal value is 2MHz
   - **Number of sweeps** indicates the number of times the entire MW frequency range has to be repeated and acquire the data for each sweep (see fig.10(a)). The more the sweep the better will be acquired data (ie lesser the noise) but greater will the entire acquisition time. So, the optimal value for the number of sweeps is 5 for this experiment.
3. After setting all the parameters, run the program and select **Sweep Mode**.(see fig.10(b))
4. Now, observe the plot panel "Normalized Photoluminescence vs Frequency". Each point in the plot corresponds to the voltage value generated by the photo-diode, due to the red fluorescence hitting its sensor, for the corresponding MW frequency. As the frequency approaches the resonance frequency the voltage decreases, which is minimum at the resonant frequency.
5. Once the acquisition of the data for all the sweeps is over and the program automatically stops, save the data by right-clicking on the ODMR plot, "Normalized Photoluminescence vs Frequency". Then go to "Export" and click on **Export to the Excel**.(see fig.10(c)). Use this data for further analysis with the help of the Origin software.

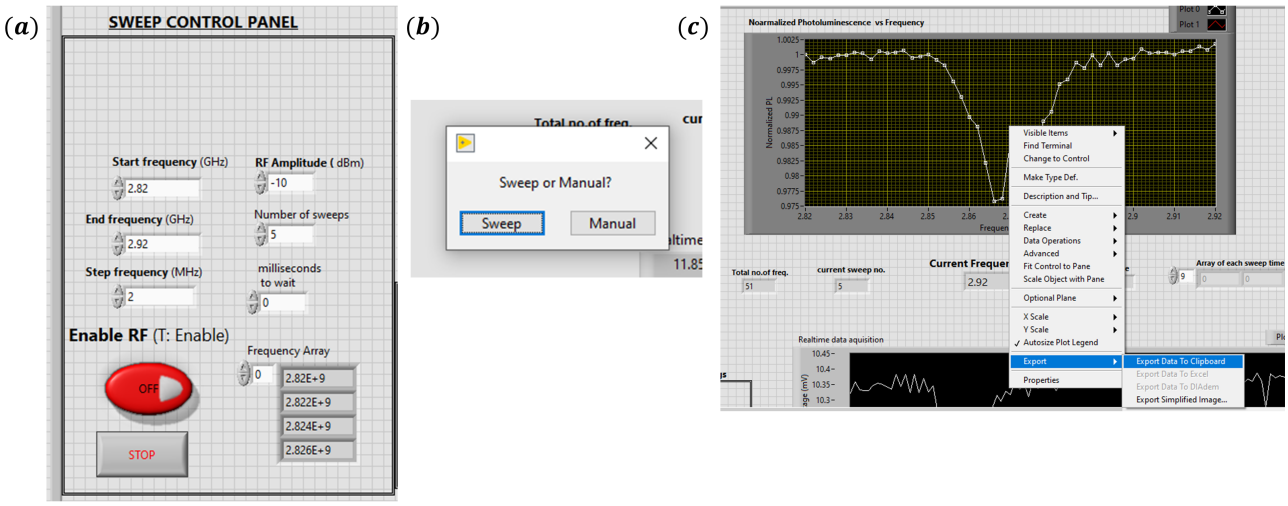


Figure 10

8.3 Origin software

1. After saving the experimental data which was exported to the Excel file, open it in the **Origin** Software.
2. Then go to the **Analysis** → **Peaks and Baseline** → **Multiple Peaks Fit**.
3. Use Lorentz function for data fitting and find the resonance frequencies from the fitting.

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